

A funoid related to directed topological spaces

OPG-60019	L1 - Tractable	Partiellement résolu	Topology
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Énoncé original (anglais, reproduction mot pour mot)

Conjecture Let R be the complete funoid corresponding to the usual topology on extended real line $[-\infty, +\infty] = \mathbb{R} \cup \{-\infty, +\infty\}$. Let $\geq$ be the order on this set. Then $R \sqcap^{\mathsf{FCD}} \mathord{\geq}$ is a complete funoid.

Proposition It is easy to prove that $\langle R \sqcap^{\mathsf{FCD}} \mathord{\geq} \rangle_{\{x\}}$ is the infinitely small right neighborhood filter of point $x \in [-\infty, +\infty]$.

If proved true, the conjecture then can be generalized to a wider class of posets.

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